

**Time:** 5 minutes**Outcome:** Use marginal productivity to analyze labor demand, wages, and employment.**Difficulty:** ●●○ Intermediate

Factor Markets & Labor Demand



THE CORE IDEA

Marginal product of labor (MPL) is the extra output from one more worker. For a price-taking seller, the value of marginal product (VMP) equals output price times MPL. A firm that also takes the wage as given hires to the point where VMP equals the wage.



HOW TO RECOGNIZE IT

- A wage change moves along labor demand; productivity or output-price changes shift it.
- Worker preferences, population, training, and alternative jobs can shift labor supply.
- Market labor demand and supply determine the competitive wage and employment.

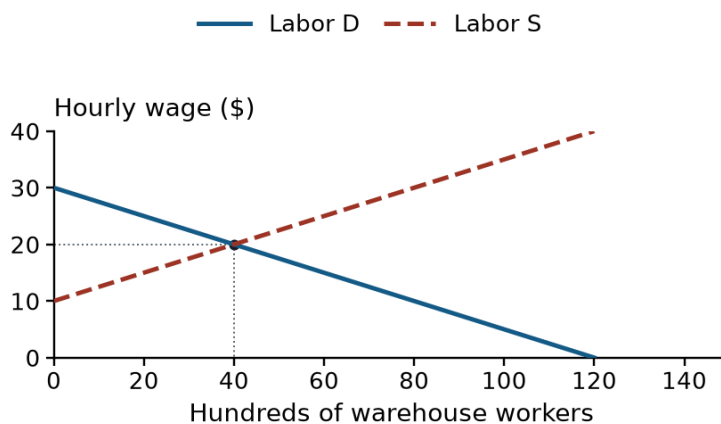


WATCH OUT

Do not confuse MPL with VMP. MPL is extra physical output; VMP converts that output into dollars by multiplying by the output price.



WORKED EXAMPLE: LABOR DEMAND AND THE WAGE



Suppose output sells for \$5 and the marginal worker adds 4 units per hour. VMP is $\$5 \times 4 = \20 per hour, matching a \$20 wage. The graph shows market equilibrium at that wage and 40 hundreds of workers, or 4,000 workers. A higher output price raises VMP at each employment level and shifts labor demand right.



CHECK YOURSELF

If the output price rises while worker productivity is unchanged, what happens to labor demand?

**READY?**

Return to the game and master this concept.